

Problem

For a two-port, let $\mathbf{A} = 4$, $\mathbf{B} = 30\ \Omega$, $\mathbf{C} = 0.1\ \text{S}$, and $\mathbf{D} = 1.5$. Calculate the input impedance $\mathbf{Z}_{in} = \mathbf{V}_1/I_1$,

when:

(a) the output terminals are short-circuited,

(b) the output port is open-circuited,

(c) the output port is terminated by a $10\text{-}\Omega$ load.

Step-by-step solution

Step 1 of 5

$$V_1 = 4V_2 - 30I_2 \quad -(1)$$

$$I_1 = 0.1V_2 - 1.5I_2 \quad -(2)$$

Step 2 of 5

(a) Out put is short circuited

$$V_2 = 0$$

$$V_1 = -30I_2$$

$$I_1 = -1.5I_2$$

$$V_1 = -30 \times \frac{I_1}{-1.5} = \frac{300}{15} I_1$$

$$\frac{V_1}{I_1} = 20\Omega$$

Step 3 of 5

(b) The Out put is open circuited

$$I_2 = 0$$

From (1) & (2)

$$V_1 = 4V_2$$

$$I_1 = 0.1V_2$$

$$I_1 = \frac{0.1V_1}{4} \Rightarrow \frac{V_1}{I_1} = 40\Omega$$

Step 4 of 5

(c) The Out put is rerminated by 10Ω Loap

$$V_2 = -10I_2$$

$$V_1 = -40I_2 - 30I_2$$

$$V_1 = -70I_2$$

$$I_1 = 0.1 \times 10I_2 - 1.5I_2$$

$$I_1 = -0.5I_2$$

$$V_1 = -70 \times \left(\frac{-I_1}{0.5} \right) = \frac{700}{5}$$

$$\frac{V_1}{I_1} = 140\Omega$$

Step 5 of 5

From (4)

$$V_2 = \frac{-g_{21}Z_L}{g_{22} + Z_L} V_1$$

$$\frac{V_2}{V_s} = \frac{-g_{21}Z_L}{g_{22} + Z_L} \times \frac{g_{22} + Z_1}{(1 + g_{11}Z_s)(g_{22} + Z_L) - g_{12}g_{21}Z_L}$$

$$\frac{V_2}{V_s} = \frac{-g_{21}Z_1}{(1 + g_{11}Z_s)(g_{22} + Z_L) - g_{12}g_{21}Z_L}$$